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Criterion A – Planning

Client Scenario

Mr. XXXXX, an experienced cricket head coach at XXXXXX XXXXX District Cricket club faces many difficulties in the coaching of bowlers. Currently, Mr XXXXX says that the only way he can provide feedback to his students is by recording and rewatching bowling footage of the student filmed from behind the stumps¹. He says that as a result of not having any system that analyses this bowling footage, there are multiple difficulties he faces in providing feedback to his junior cricketers, these issues include:

- 1) Lack of being able to review bowling trajectory.²
- 2) Lack of being able to accurately measure line & length of deliveries.³
- 3) Lack of being able to store bowling performance analysis to share with his students.⁴

These issues pose difficulties for **Mr XXXXX** that hinder him from providing effective and insightful feedback to his students, which inhibits the bowling development of these students.⁵ Seeing this as an opportunity for my Computer Science IA, I offered to build an application to cater for his needs. He accepted my proposal, and we scheduled a meeting to gather his requirements in depth.

Rationale

As my proposed solution, I want to create a desktop application that will be stored locally on Mr. XXXXX's computer. The backend will be coded in Flask using Python, and the user can navigate the front end through multiple HTML webpages. The application will allow Mr XXXXX to upload bowling videos of his students, and these videos will act as input into the application. These videos will be analysed and processed by the application based on which Mr XXXXX can receive analysis features such as ball tracking and statistics on the pitching/landing area of the delivery about the submitted video. These features will be displayed on the webpage for Mr XXXXX to gain insightful analysis of the bowling performance of their student in a more objective way, allowing him to provide more effective feedback to his students.

For video processing, detecting the ball and stumps within the videos will be necessary to develop these features. To do this, I plan to annotate bowling videos from diverse filming environments using an open-source CVAT (Computer Vision Annotation Tool) software. Following this, I will use this labelled dataset to train a Yolov5 object detection machine learning model. Even though there are several ways of implementing this object detection, Yolov5's fast and efficient inference and simple integration in Python (with libraries such as Pytorch, OpenCV2 and Numpy) makes it the most optimal method of object detection for this product. These libraries can seamlessly integrate the Yolov5 Object detection outputs, such as coordinates/position of objects within an image, with other computer vision features of Python libraries, such as video conversion into frames, image editing and video compilation.

¹ Interview 1: Appendix A

² Interview 1: Appendix A

³ Interview 1: Appendix A

⁴ Interview 1: Appendix A

⁵ Interview 1: Appendix A

Finally, for the front end of the web application, I plan to use a web application framework written in Python called Flask. Flask can provide me with tools that can allow me to seamlessly link the Python computer vision processing with a web interface. For the web interface, I will create multiple webpages (IIS pages) for uploading videos (input) and displaying analysis (output) including ball-tracked video and ball landing images. With the Jinja Templates engine in Flask, I will link my server-side scripting in Flask with easily navigable UI with HTML, CSS and Javascript.

Success criteria

1: Video Viewing	Coaches should be able to select a video and view it in the webpage with the ability to slow the speed for scrutiny of the video prior to submission.
2: Video Submission	Coaches should be able to upload a video up to 5 seconds in length, of each delivery bowled by the student to the application.
3: Ball Trajectory	The application should be able to produce an output video that clearly shows the trajectory path of the ball using a visible white line.
4: Length/Line Visual	The output video should clearly show the spot where the ball bounces, highlighted by a red dot within the output video.
5: Length/Line Calculation	The application should be able to determine how many meters the ball bounces away from the stumps of the striker.
6: Length/Line Analysis	The application should be able to produce a pitch map graphic on which the bounce spot is shown by a dot drawn at the bounce position of the specific delivery in the submitted video.
7: Line/Length Master	The application should be able to produce a pitch map that displays the bounce spot of all deliveries that have been uploaded to the application till date.

Word Count: 498

